I. UNDERSTANDING ICT IN EDUCATION

Information and Communications Technology (ICT)

ICT is any technology that can be used to access and send information, and to connect to other devices via internet, wireless network, or other communication media.

One of the examples of ICT is computer. It is a programmable device used to execute arithmetic and logical operations. There are two (2) parts of a computer:

1. **Hardware.** It is the physical structure of the computer; thus, it is tangible. There are four (4) categories of hardware:

   - **Input Devices.** These are hardware devices that allow the users to enter data into the computer. Examples of these are the following:

     - **Keyboard.** This allows you to input various characters into the computers, such as letters, numbers, and symbols. It is also used to execute commands (or also known as keyboard shortcuts) in the computer.

Command	Description
Ctrl + A	Selects all text in the current document
Ctrl + B	Transforms the text into boldface font
Ctrl + C	Copies an object or text
Ctrl + D	Displays the font dialogue box
Ctrl + E	Changes a paragraph between center and left alignment
Ctrl + F	Displays the Find dialogue box
Ctrl + G	Displays Go To dialogue box
Ctrl + H	Displays Replace dialogue box
Ctrl + I	Italicizes text
Ctrl + J	Changes paragraph from Justified to Left alignment
Ctrl + K	Creates hyperlink
Ctrl + L	Aligns the paragraph into left
Ctrl + M	Indents the paragraph from the left
Ctrl + N	Creates a new document
Ctrl + O	Opens a new document
Ctrl + P	Prints document
Ctrl + R	Changes paragraph alignment from left to right
Ctrl + S	Saves the document
Ctrl + U	Underlines text
Ctrl + V	Pastes the copied item or text
Ctrl + X	Cuts the selected item or text
Ctrl + Y	Redo the last section
Ctrl + Z	Undo the last section

- **Mouse.** It is a device used to control the cursor and has the capability to select texts, icons, and other objects in the graphic user interface (GUI).

- **Output Devices.** These are hardware devices that allow the computer to communicate with its users and other devices. It also receives data from the computer for display and physical reproduction. Examples are the following:

  - **Monitor.** It is an output device that displays texts and graphics in the computer.

  - **Printer.** It is a peripheral device that allows the user to reproduce the data in the computer onto a physical medium like paper. There are various types of computers namely:

    a. Dot Matrix Printer. It is type of printer that prints text or image by striking an ink ribbon and form dots.

    b. Inkjet Printer. It is a type of printer that prints text and images by spraying streams of quick drying ink on the paper.

    c. Laser Printer. It is a type of printer that uses laser technology to produce images in the paper.

    d. 3D Printer. It is the new trend in printing technology that uses metal alloys, polymers, plastics, and food ingredients to create a physical object from a digital model.

  - **Projector.** It is an output device that uses a light to display an image on a flat surface.

  - **Speakers.** It is an output device that produces sound that is made by the computer.

- **Storage Devices**

  - **Hard Disk Drive.** It is the major storage device of a computer. The operating system, data, and all files in the computer are stored in a hard disk drive.

  - **USB Flash Drive.** It is a thumb-sized and portable storage device used to save files and media.

- **Processing Devices**

  - **Central Processing Unit (CPU).** It is a device that processes all instructions received from all the hardware and softwares running on the computer. It is also known as the brain of the computer.

  - **Motherboard**. It is a printed circuit board (PCB) that contains the CPU, Random Access Memory (RAM), and all other computer hardware components.

2. **Software.** This is the intangible part of the computer. It is the collection of instructions that allow the users to interact with a computer. There are two categories of softwares:

- **Operating System.** This is a software program that allows the hardware to execute instructions from the software. Examples of OS are Windows, MacOS, Linux, and Android.
- **Application Software.** It is also known as the end- user programs. These are the softwares that have specific functions needed by a user to accomplish a specific task. Examples of application softwares are the following:
  - Word Processing
  - Spreadsheets
  - Presentation software

Educational Technology

It includes any media that can be used in instruction (Borjal 2013). Educational Technology may also be defined using the traditional versus the progressive perspective:

- From the perspective of ***Traditionalists***, educational technologies are learning tools that function as an alternative to teachers (**Learning FROM technology**). Example:

  Instead of enrolling in a vocational program, aspiring bakers can now search relevant videos in an online video streaming site to learn how to bake.

- Meanwhile, Progressivists believe that educational technologies are learning tools that enhance the teaching and learning process (**Learning WITH technology**). Example:

  Using of simulation tool such as Virtual Reality box to teach the Solar System

Roles of ICT in Education

According to Dr. Borjal (2013), there are two roles of ICT, specifically the computer, in teaching and learning:

- *Computer as Object of Instruction.* This pertains to the role of computer to education as subject matter. This refers to computer literacy—it is gaining elementary to advanced knowledge and experience from using computers.

  Example: Studying computer programming as a subject in BS Information Technology
- *Computer as Tool for Instruction.* This refers to the role of computer in education as a device in enhancing teaching and learning.

  Example: Utilization of mobile app in teaching anatomy

Distance Education

- Distance learning can be described as a method of education that is received by a learner at another geographical location. There are two types of distance education:
  - **Synchronous distance learning.** It is learning that involves live communication through either chatting online, sitting in a classroom, or

even teleconferencing. It is one of the most acclaimed distance learning types that is most suitable for engaging in continuing education programs.

- **Asynchronous distance learning.** It is a type of learning that has a strict set of deadlines, often a weekly time limit; however, it allows learners to learn at their own pace. It is also one of the most popular distance learning types because students can communicate with each other seamlessly through online notice/bulletin boards.

Massive Open Online Courses (MOOC)
According to Kurt (2018), MOOC are asynchronous, open-access, Web-based courses geared toward enrolling hundreds or thousands of students at a time. MOOCs deliver content via recorded video lectures, online readings, and online assessments, as well as various degrees of student-student and student-instructor interaction.
The following are the well-known MOOCs:

Website Name	URL
Coursera	https://www.coursera.org/
edX	https://www.edx.org/
Udacity	https://www.udacity.com/

II. ICT POLICIES AND SAFETY ISSUES IN TEACHING AND LEARNING
1. Republic Act 10844—Department of Information and Communications Technology Act of 2015

- This was signed last May 23, 2016.
- It is a law that created the DICT, an administrative entity of the Executive branch that is responsible in planning, developing, and promoting the national ICT agenda in the Philippines.

2. ICT4E Strategic Plan (Information and Communications Technology for Education)

- It is a 5- year plan by Department of Education that aims to develop an ICT- enabled curriculum. There are two stages of the curriculum reform:
  - Enhancing ICT use in existing curriculum; and
  - Full integration of ICT in enriched curriculum.
- There are different steps in the implementation of the ICT4E, namely:
  - **Step 1 (first three years)**
    - ICT-enhanced curriculum
    - ICT resources
    - Research, monitoring, and evaluation of impact of ICT
  - **Step 2 (last two years)**
    - ICT-enriched curriculum
    - Mainstreamed ICT models
    - New delivery systems

- ICT4E also aims to lead the students to be proficient, adaptable lifelong learners where ICT plays a major role in creating new and improved model of teaching and learning where education happens anytime and anywhere. Learning targets for students have been identified in the following six areas:
    - Basic operations and concepts
    - Social, ethical, and human issues
    - ICT for producing
    - ICT for communicating
    - ICT for researching
    - ICT for problem-solving

III. Theories and Principles in Educational Technology

1. Dale's Cone of Experience

- It is a visual model that is composed of ten (10) stages, starting from the concrete experiences at the bottom of the cone, to more abstract experiences as it reaches the peak of the cone.
- Concrete learning is from first- hand experiences, and it incorporates the use of all five senses. Conversely, abstract learning is from intuition and imagination.

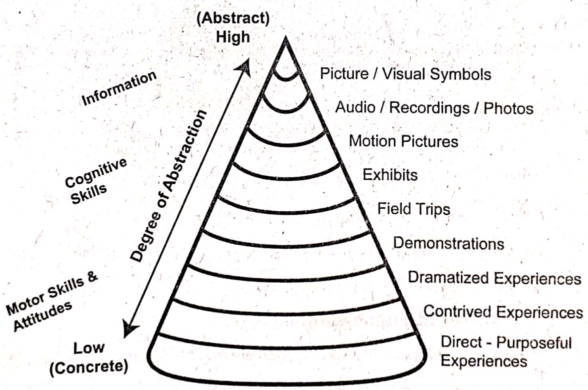
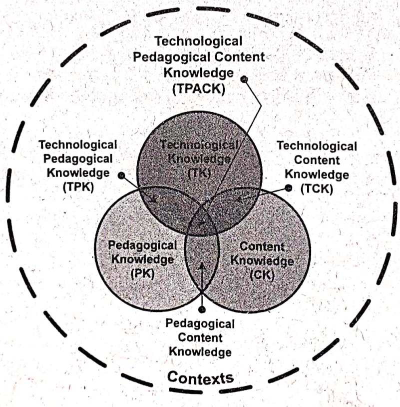


Figure 1: Dale's Cole of Experience

2. Three-Fold Analysis of Experience by Jerome Bruner

- He suggests that every area of knowledge that is included in the cone of experience can be presented and learned in three district steps, namely:
    - **Enactive.** This refers to direct experiences. In Dale's Cone of Experience, it pertains to the following:

  - Direct, purposeful experiences
  - Contrived experiences
  - Dramatized experiences
- **Iconic.** This pertains to pictorial experiences. It refers to the following:
  - Demonstrations
  - Field trips
  - Exhibits
  - Motion pictures
  - Recordings and still pictures
- **Symbolic.** These are the highly abstract experiences.
  - Visual Symbols
  - Verbal Symbols



3. TPACK Framework

Figure 2: TPACK Framework by Matthew Koehler & Punya Mishra

- It is a framework that was proposed by Matthew J. Koehler and Punya Mishra of Michigan State University. They identified the three primary forms of knowledge:
  - **Content Knowledge.** It pertains to the specialization of the teacher. This means that the teacher should be an expert and should have a solid understanding of what he/she teaches.
  - **Pedagogical Knowledge.** It is the teacher's advanced knowledge about the strategies, methods, and processes in teaching and learning.

- **Technological Knowledge.** According to the Committee of Information Technology Literacy of the National Research Council (as cited by Koehler & Mishra, 2006), FITness (Fluency of Information Technology) requires deeper, more essential understanding and mastery of information technology for information processing, communication, and problem-solving.

- In Figure 2, each of the primary forms of knowledge creates intersections and relationships which will be explained below.
  - **Technological Content Knowledge**. In here, teachers should know the appropriate and best technology in teaching and learning a specific subject matter.
  - **Technological Pedagogical Knowledge.** This is a type of knowledge in which teachers should know what suitable technology can enhance teaching and learning experiences.
  - **Pedagogical Content Knowledge.** This pertains to the knowledge in which teachers should know the best practices in teaching specific content to specific students.

4. Gagne's 9 Instructional Events

- This was proposed by Robert Gagne, who proposed a systematic instructional design process that shares the behaviorist approach to learning.
- According to Gagne, Briggs, and Wager (as cited by the Northern Illinois University, Faculty Development and Instructional Design Center, n.d.), the following are the nine steps in instruction:
  - Gain attention of the students
  - Inform students of the objectives
  - Stimulate recall of prior learning
  - Present content
  - Provide learning guidance
  - Elicit performance (practice)
  - Provide feedback
  - Assess performance
  - Enhance retention and transfer to the job

5. Merill's Principles of Instruction

- According to Merill (2002), the first principle relates to problem-centered instruction. Four more principles are stated for each of the four phases for effective instruction. These five first principles, stated in their most concise form, are as follows:
  - Learning is promoted when learners are engaged in solving real-world **problems** — start with simple problems and work through a progression of increasingly complex problems.

- Learning is promoted when existing knowledge is **activated** as a foundation for new knowledge — prior experience from relevant past experience is used as a foundation for the new skills and knowledge (also known as scaffolding).
- Learning is promoted when new knowledge is **demonstrated** to the learner — they are shown, rather than just being told.
- Learning is promoted when new knowledge is **applied** by the learner — they are required to use their new knowledge or skill to solve problems.
- Learning is promoted when new knowledge is **integrated** into the learner's world — they are able to demonstrate improvement in their newly acquired skills and to modify it for use in their daily work.

6. Levels of Technology Integration

- There is a technology integration because technological advancement in inevitable. The **Technology Integration Matrix (TIM)** provides framework for describing and targeting the use of technology to enhance learning.

→ LEVELS OF TECHNOLOGY INTEGRATION ↓ CHARACTERISTICS OF THE LEARNING ENVIRONMENT	ENTRY LEVEL The teacher begins to use technology tools to deliver curriculum content to students.	ADOPTION LEVEL The teacher directs students in the conventional and procedural use of technology tools.	ADAPTATION LEVEL The teacher facilitates students in exploring and independently using technology tools.	INFUSION LEVEL The teacher provides the learning context and the students choose the technology tools to achieve the outcome.	TRANSFORMATION LEVEL The teacher encourages the innovative use of technology tools. Technology tools are used to facilitate higher order learning activities that may not have been possible without the use of technology.
ACTIVE LEARNING Students are actively engaged in using technology as a tool rather than passively receiving information from the technology.	Information passively received	Conventional, procedural use of tools	Conventional independent use of tools; some student choice and exploration	Choice of tools and regular, self-directed use	Extensive and unconventional use of tools
COLLABORATIVE LEARNING Students use technology tools to collaborate with others rather than working individually at all times.	Individual student use of tools	Collaborative use of tools in conventional ways	Collaborative use of tools; some student choice and exploration	Choice of tools and regular use for collaboration	Collaboration with peers and outside resources in ways not possible without technology
CONSTRUCTIVE LEARNING Students use technology tools to connect new information to their prior knowledge rather than to passively receive information.	Information delivered to students	Guided, conventional use for building knowledge	Independent use for building knowledge; some student choice and exploration	Choice and regular use for building knowledge	Extensive and unconventional use of technology tools to build knowledge
AUTHENTIC LEARNING Students use technology tools to link learning activities to the world beyond the instructional setting rather than working on decontextualized assignments.	Use unrelated to the world outside of the instructional setting	Guided use in activities with some meaningful context	Independent use in activities connected to students' lives; some student choice and exploration	Choice of tools and regular use in meaningful activities	Innovative use for higher order learning activities in a local or global context
GOAL-DIRECTED LEARNING Students use technology tools to set goals, plan activities, monitor progress, and evaluate results rather than simply completing assignments without reflection.	Directions given; step-by-step task monitoring	Conventional and procedural use of tools to plan or monitor	Purposeful use of tools to plan and monitor; some student choice and exploration	Flexible and seamless use of tools to plan and monitor	Extensive and higher order use of tools to plan and monitor

The Technology Integration Matrix was developed by the Florida Center for Instructional Technology at the University of South Florida, College of Education. For more information, example videos, and related professional development resources, visit http://mytechmatrix.org This page may be reproduced by districts and schools for professional development and pre-service instruction.

Figure 3: Technology Integration Matrix
Source: https://eltplanning.files.wordpress.com/2017/10/tech2.png

IV. Instructional Design Models

1. ADDIE Model

- It is an instructional design model that serve as a method in planning and developing educational and training programs.

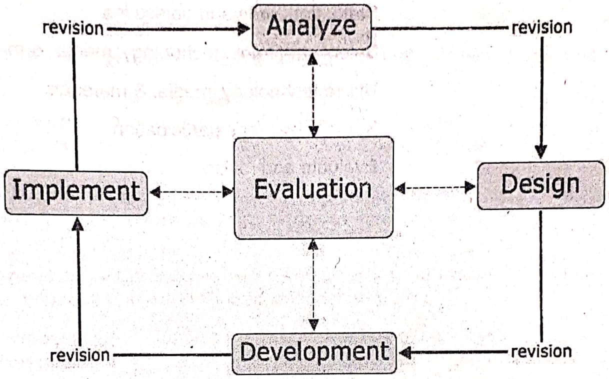


Figure 4: ADDIE Instructional Model
Source: https://educationaltechnology.net/the-addie-model-instructional-design/

- **ADDIE** has the following stages:
    - **Analyze.** This is where the goal is set. In this phase, the focus of the teacher is the students. This is where the prior skills and knowledge of the students are being analyzed. This is to ensure that the program is suitable to level of their skills and knowledge.
    - **Design.** The design phase deals with learning objectives, assessment instruments, exercises, content, subject matter analysis, lesson planning, and media selection. The design phase should be systematic and specific.
    - **Development.** In this phase, the designer assembles and organizes the objective, content, assessment and others that have been set in the design phase.
    - **Implement.** In this stage, the program that has been developed should be executed to the students.
    - **Evaluation.** This phase consists of two types of assessment: the formative and summative assessment. This is to test if the goals that has been set were achieved.

2. **ASSURE model**

- This is an instructional systems design (ISD) that aims to design and develop the most appropriate learning experiences and environment for the students.
- It has the following phases:
    - **A**nalyze learners
    - **S**tate standards and objectives
    - **S**elect strategies, technology, media, & materials
    - **U**tilize technology, media, & materials
    - **R**equire learner's participation
    - **E**valuate and revise

1. Which is the most critical consideration when choosing an instructional material?

   A. Learning content
   B. Learning outcomes
   C. Durability
   D. Aesthetic appeal

The answer is letter B. According to the systematic approach to instruction, defining the learning outcomes or objectives shall be the most critical consideration when choosing an instructional material.

2. Teacher Max wants to teach his students about poisonous snakes. Which is the best experience to employ?

   A. Iconic experiences
   B. Enactive experiences
   C. Symbolic experiences
   D. Actual experiences

The answer is letter A. Since snakes are one of the dangerous animals, iconic or pictorial experience is the best to employ.

3. Ms. Lea wishes to perform literary analysis on the novel "To the Lighthouse." Which of the following is the most appropriate level of experience for this?

   A. Symbolic experiences
   B. Iconic experiences
   C. Enactive experiences
   D. Experiential experiences

The answer is letter A. Literary analysis deals with visual symbols. Hence, the suitable experience to employ is symbolic.

4. This type of mode of distance learning involves providing audio-video media to large audiences.

   A. Simultaneous learning
   B. One - way video
   C. Video - conferencing
   D. Virtual classrooms

The answer is letter B. It is type of audio- video media that can be sent to multiple sites.

5. Which among these terms is the most encompassing in the context of applying technology in the teaching-learning process?

   A. Information Technology
   B. Instructional Technology
   C. Educational Technology
   D. Technology in Education

The answer is letter C. EdTech is the application of technology in enhancing teaching and learning.

6. As a technology applied in education, which is the most limited in terms of the amount of information that may be transmitted through its use?

   A. chalkboard
   B. graphs & charts
   C. Internet
   D. moving film

The answer is letter A. In Dale's cone of experience, text symbols are the least among the other experiences.

7. Which statement supports the ultimate goal of integrating the use of technology in education?

   A. Learning happens better outside the classroom.
   B. Technological advancement is inevitable.
   C. Teachers are needed less in learning.
   D. Technology must penetrate all aspects of education.

The answer is letter B. This suggests that education is dynamic, and it should adapt the utilization of technology because its advancement is unavoidable.

8. Nonato cannot attend his class because of chicken pox and cannot attend school for the meantime. The teacher decided to provide him with a kit of activities similar to his classmates so he won't be left behind. This program is also known as ___________.

   A. Alternative Learning Systems
   B. Distance Learning
   C. Informal Education
   D. Modular Lesson

The answer is letter D. Nonato needs a material that is self-sufficient since he will be studying at home. The most suitable material that his teacher may provide him is a module. It is a unit of work in a course of instruction that is virtually self-contained and a method of teaching that is based on the building up skills and knowledge in discrete units.

9. The purpose of the instructional materials are the following except?

   A. Make abstract concept more concrete.
   B. Concept and easily absorbed.
   C. Effective instruction and effective cost.
   D. Entertainment value.

The answer is letter A. The primary purpose of instructional materials is to enhance teaching and learning by providing concrete experiences rather than purely abstract learning.

10. Teacher Mel brought a white mouse in the class during a discussion about mammals. The white mouse is a device commonly known as realia. Teacher can bring realia only when it is________.

   A. available
   B. feasible
   C. affordable
   D. workable

The answer is B. Realia are objects from real life that is used as an instructional material. Realia can be brought inside the classroom only when it is possible or feasible.

11. Teacher John is using technology to enhance the learning in his subject matter. He is practicing:

    A. gamification
    B. collaboration
    C. technology - driven
    D. integration

    The answer is letter D. Integration or Technology integration is the application of technology in teaching and learning process.

12. The following are the instructional materials as supplements to textbooks, except:

    A. computer software
    B. tables, charts, and maps
    C. guide questions
    D. mobile app

    The answer is letter C. The other choices are used to enhance or supplement textbooks, but guide questions are used to provide focus and direction in studying the textbooks.

13. Teacher Gabriel has shown only the important portion rather than whole video as a motivation to his lesson in the class. What did he achieve?

    A. saving on electricity
    B. avoiding unnecessary noise
    C. better use of time
    D. managing discipline

    The answer is letter C. If teacher Gabriel have shown only the important portion of the video, he may have time left for reflections and other relevant activities.

14. The following are the importance of internet and world wide web in the teaching and learning, except:

    A. Access to a large database of information
    B. Support open distance learning
    C. Extended learning opportunities
    D. Can be used in search of a lifetime partner

    The answer is letter D. Finding a lifetime partner via internet and world wide web has nothing to do with teaching and learning.

15. Arrange chronologically the following events of instruction that was developed by Robert Gagne:

    I. Elicit performance
    II. Provide feedback
    III. Gain attention
    IV. Inform learners of objectives
    V. Stimulate recall of prior learning

    A. I, III, II, and V
    B. II, V, III, and I
    C. II, V, III, IV, and I
    D. III, IV, V, I, and II

The answer is letter D. The 9 events of instruction by Robert Gagne follows these steps: (1) Gain attention of the students, (2) Inform learners of the objectives, (3) Stimulate recall of prior learning, (4) Present the content, (5) Provide learning guidance, (6) Elicit performance, (7) Provide feedback, (8) Assess performance, (9) Enhance retention and transfer to the job.

16. In providing the students access to the internet, teachers should observe these standards, except:

    A. Teach students to test reliability and authenticity of a website.
    B. Allow students to independently explore the world wide web.
    C. Require students to cite all the websites they use to ensure that they give proper credit and avoid plagiarism.
    D. Inform students of their limitations in accessing the internet.

The answer is letter B. Teachers should inform and monitor the students of their boundaries in accessing the internet.

17. Authentic learning can also be similar with this level in Dale's Cone of Experience.

    A. contrived experiences
    B. direct, purposeful experiences
    C. dramatized experiences
    D. field trips

The answer is letter B. Direct, purposeful experiences is same as authentic learning. It refers to gaining knowledge and skills through real-life and first-hand experiences.

18. What level of technology integration is reached by Teacher Liza who builds a learning environment that instills the power of technology tools throughout the day and across subject areas?

    A. Infusion
    B. adaptation
    C. transformation
    D. adoption

The answer is letter C. Transformation is a level of technology integration wherein the class would be impossible to happen without the technology. In the case of Teacher Liza, her goal is to instill the power of technology tools to students and it would not be possible to achieve without the technology.

19. What educational technology is most appropriate in showing the trend in temperature change from month-to-month?

    A. bar graph
    B. flowchart
    C. drawing
    D. map

The answer is letter A. Among the choices, the suitable educational technology to be used in showing the trend in temperature change from month to month is bar graph.

20. Blended learning is made possible partly through actual classroom meetings coupled with ______ instruction.

A. print module
B. tape- recorder
C. online
D. private tutor

The answer is letter C. Blended learning is an education program that combines online media and traditional classroom setting.

21. Among the five levels of technology integration, what is achieved by Teacher Ruby who directs students on the conventional use of tool-based software like Microsoft as an entry point?

A. adaptation
B. infusion
C. adoption
D. transformation

The answer is letter C. Adoption is a level of technology integration when the teacher directs the students in the conventional and procedural use of technology tools.

22. Teacher Toni builds a rich learning environment in which students regularly engage in activities that would have been impossible to achieve without technology, e.g their own PowerPoint presentation of lessons. This act of Teacher Toni can be classified to this level of technology integration.

A. adoption
B. adaptation
C. infusion
D. transformation

The answer is letter D. Transformation is a level of technology integration in which the technology tools are used to facilitate higher-order learning activities that may not have been possible without the use of technology.

23. What level of technology integration is achieved by Teacher Fe who applies collaborative learning and also enables students to use technology to collaborate with peers and experts regardless of time zone and physical distances, etc.?

A. infusion
B. entry
C. transformation
D. adaptation

The answer is letter C. In this question, Teacher Fe conducts distance learning. It is a method of education that is received by the learner at another geographical location. Distance education will not be possible to be conducted without technology. Therefore, distance education is an example of Transformation.

24. In ICT, what component advanced the application of modern-day digital technology (PC's phones, internet communication, etc.) that was superior to the analogue technology in the late 1900's (e.g. movies using projectors, line telephone communication, beepers, etc.)?

A. electronic circuit
B. physical structure
C. editorial content
D. entertainment content

The answer is letter B. The difference between the modern- day digital technology and the analogue technology in the late 1900s is physical structure—from analogue to digital.

25. MOOC's or Massive Open Online Courses are open since they are ____________.

    A. free for all
    B. in the internet
    C. available during the opening of the school year
    D. open to ideas without limit

The answer is letter B. It is open since it is accessible anytime and anywhere via internet.

26. With the influx of technology in the 21st century, how can students be best re-educated on the ethical use of social media, such as Facebook and mobile texting?

    A. Integrate social media competencies in the curriculum
    B. Students monitoring of peers
    C. Punishment for violators
    D. Seminar on social media

The answer is letter A. The most feasible action that the teachers can do to re-educate students on the ethical use of social media is through integrating social media competencies in the curriculum.

27. What is the appropriate response for millennials who are more exposed to virtual reality (digital toys and tools) instead of actual reality?

    A. No mobile phones in school
    B. Limits use of Facebook and texting
    C. PTA conference with parents
    D. Align learning in context of reality

The answer is letter D. Virtual reality can be relevant in education if the teacher aligns learning in the context of reality.

28. Clear and organized presentation for lecture and/or instruction can be done by means of________.

    A. PowerPoint
    B. diskette
    C. acetate
    D. glass board

The answer is letter A. Among the choices, the most clear and organized presentation can be done thru PowerPoint.

29. Face-to-face communication is made possible via computer through the use of______.

A. Email
B. Twitter
C. Skype
D. Google

The answer is letter C. Skype is a computer app that is used specifically for online chatting and face-to-face communication.

30. Internet is the term used to describe the ______.

A. software
B. information super highway
C. desktop computer
D. cellphone

The answer is letter B. In 1900s, internet was being called as "information superhighway."

31. According to Republic Act 10175 or otherwise known as "Cybercrime Prevention Act of 2012," the following are forms of cybercrime, except:

A. illegal logging
B. identity theft
C. cybersex
D. illegal access

The answer is letter A. Logging in and out are terms that can be incorporated with the cyber world, but illegal logging is a form of deforestation.

32. He is called as the "Father of Computer" for inventing the Analytical Engine.

A. Charles Babbage
B. Blaise Pascal
C. Tim Berners- Lee
D. Mark Zuckerberg

The answer is letter A. Charles Babbage invented the first computer, the Analytical Engine, which has made him the "Father of Computer."

33. Microsoft Company is popular when it comes to software manufacturing. Who introduced the Microsoft?

A. Steve Jobs
B. Tim Berners-Lee
C. Mark Zuckerberg
D. Bill Gates

The answer is letter D. Bill Gates is an American business magnate, investor, author, philanthropist, humanitarian, and principal founder of Microsoft Corporation.

34. Teacher Mel ensures that the mobile app she intends to utilize in teaching history develops the standard competencies prescribed by curriculum guide. Which of the following factors she is considering?

A. objectivity
B. coverage
C. accuracy
D. competence

The answer is letter A. Evaluating. Evaluating if the mobile app will develop to the learners the standard competencies prescribed by the curriculum guide means checking its objectivity.

35. Teacher Jen allowed her students to conduct survey through an online form with different devices. What has been part of ICT in her activity?

A. ICT for producing
B. ICT for communicating
C. ICT for researching
D. ICT for problem- solving

The answer is letter C. Conducting survey online pertains to the role of ICT in research.

36. Republic Act 10844 is also known as ____________.

A. The Enhanced Basic Education Curriculum Act of 2013
B. Department of Information and Communication Technology Act of 2015
C. ICT Act of 2012
D. K to 12 Act of 2013

The answer is letter B. Republic Act 10844 is also known as Department of Information and Communication Technology Act of 2015. It creates and mandates DICT as the primary policy, planning, coordinating, implementing, and administrative entity of the Executive Branch of the government that will plan, develop, and promote national ICT development agenda.

37. This is the goal-setting stage in ADDIE instructional model.

A. Design
B. Development
C. Analysis
D. Evaluation

The answer is letter C. In the Analysis phase, the designer should define the goals or the objectives of the program.

38. It is the set of rules about the behavior acceptable on the internet.

A. Code of Ethics
B. Netiquette
C. Law of Internet
D. R.A. 10844

The answer is letter B. Netiquette is the portmanteau of network or internet and etiquette. It is the acceptable behavior online.

39. This following apps can be used for collaborative work, except ______________:

A. Google Drive
B. Edmodo
C. Wikispaces
D. Google Chrome

The answer is letter D. All of the choices are can be used for collaborative work except Google Chrome—it is a web browser that is used to display websites and webpages.